



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE19CS352 - Cloud Computing

Total Marks : 100.0

1.a. Explain the following terms with respect to Cloud Computing

- a. Elasticity
- b. Rapid provisioning
- c. Measured Service
- d. Resource Pooling

(4.0 Marks)

1.b. List and explain the benefits of moving applications to the cloud.

(4.0 Marks)

1.c. Describe briefly 3 service models and 3 deployment models of cloud computing.

(6.0 Marks)

1.d. Explain different degrees of parallelism with an example (6.0 Marks)

2.a. Consider a situation where you are required to apply any one of these types of virtualization

Full Virtualization, Bare Metal virtualization, Host based virtualization and Para Virtualization

to different implementation technologies. Mark the appropriate virtualization type for each requirement and justify your answer.

(i) Run some dedicated applications on the VMs created on the guest OS and run

some other applications on the host OS directly

(ii) Run special APIs requiring substantial OS modifications in a VM

(iii) Run non-critical instructions on the hardware directly while critical instructions are discovered and replaced with traps into the VMM to be emulated by software.

(iv) Install the virtualization software directly on the hardware (8.0 Marks)

2.b. How paravirtualization and full virtualization techniques are used to virtualize x86? (5.0 Marks)

2.c. What are controller-manager, kubelets and pods in Kubernetes? Explain with a diagram where each of them execute – on master or worker? (5.0 Marks)

2.d. List one similarity and one difference between Docker container and a VM (2.0 Marks)

3.a. Explain (5M) Lustre file system architecture with a neat diagram (3M). (8.0 Marks)

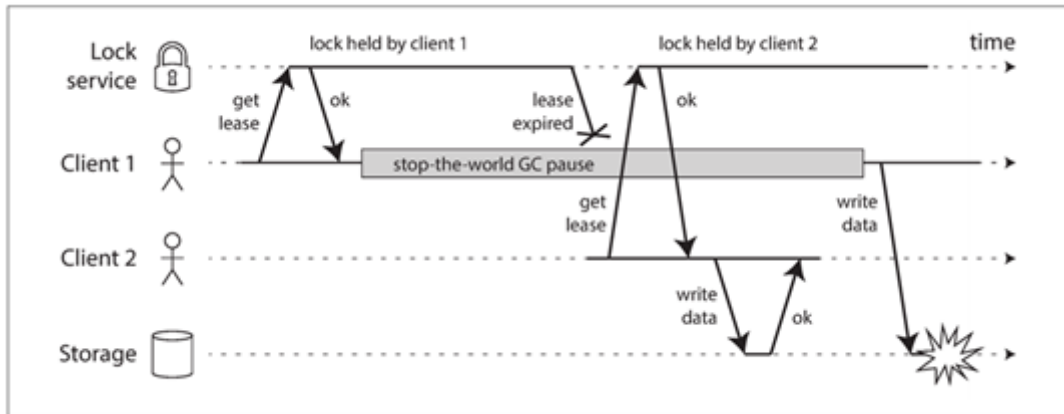
3.b. Briefly explain CAP theorem? (2 M) Explain how would you choose a database for your application based on CAP theorem (3M) (5.0 Marks)

3.c. What is a consistency model? Explain briefly any 4 types of consistency models. (5.0 Marks)

3.d. What is rebalancing of partitions? Why is it necessary to rebalance partitions? (2.0 Marks)

4.a. What is the purpose of Leader Election in Distributed computing? (2)
Explain briefly Bully Algorithm and Leader election in a Ring (6) (8.0 Marks)

4.b. What is the problem with the implementation of a distributed lock in the following diagram? Explain with a diagram the approach that is used to overcome the problem.



(6.0 Marks)

4.c. How does Zookeeper work? What are the common services offered by Zookeeper ?

(6.0 Marks)

5.a. Explain the following types of design pattern

1. Defense in Depth
2. Honeypots
3. Sandboxes
4. Network Patterns

(8.0 Marks)

5.b. What is Cloud Bursting? Explain how Cloud Bursting can be Beneficial to Cloud Users. (6.0 Marks)

5.c. What is multi-tenancy and mention its benefits in Cloud Computing. (2 Marks)

You are asked to design a multitenant database for two universities – **HighTechUniv** and **GlobalUniv** to store information about students. *HighTechUniv* wants to store USN, student names and email ids while *GlobalUniv* wants to store USN, student names and grades. Design a multitenant database using the preallocated column method for the same. (4 Marks) (6.0 Marks)